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COURSE CODE : CSA0311  
COURSE : Data Structure.

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CO<sub>2</sub>- Assessment Tool - 2

Q1] Convert the following infix expression into a postfix expression

$B - ((C/D + (E * F * G) / H) * J)$ . Also, evaluate the given postfix expressions a)  $9384/*+$  b)  $62+3$

| Reading element | Stack element | Output          |
|-----------------|---------------|-----------------|
| B               | empty         | B               |
| -               | -             | B               |
| (               | -(            | B               |
| C               | -(            | BC              |
| /               | -(/           | BC              |
| D               | -(/           | BCD             |
| +               | -(+/          | BCD/            |
| (               | -(+(          | BCD/            |
| E               | -(+(          | BCD/E           |
| +               | -(+(/         | BCD/E           |
| F               | -(+(/         | BCD/EF          |
| *               | -(+(*         | BCD/EF*         |
| G               | -(+(*         | BCD/EF*G        |
| )               | -(+           | BCD/EF*G        |
| H               | -(+/          | BCD/EF*G/H      |
| )               | -             | BCD/EF*G/H/     |
| *               | -*            | BCD/EF*G/H/     |
| J               | -*            | BCD/EF*G/H/J    |
| End             | Empty         | BCD/EF*G/H/J* - |

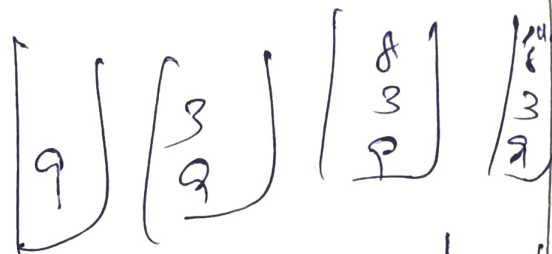
a) 9 3 8 4 / \* +



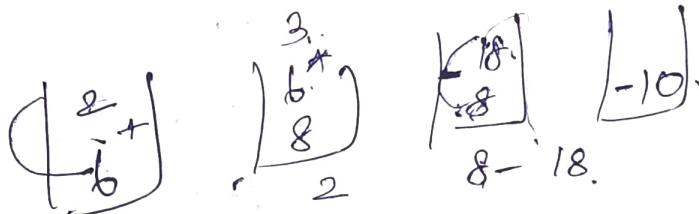
4  
8  
3  
9

$\Rightarrow$ 

|   |   |               |    |
|---|---|---------------|----|
| + | 6 | $\Rightarrow$ | 15 |
| * | 9 |               |    |
| 2 |   |               |    |
| 3 |   |               |    |
| 9 |   |               |    |



b) 6 2 + 6 3 \* -



Q2] Illustrate the queue operation using following function call of size = 5. Enqueue(2), Enqueue(67), Enqueue(89), Dequeue(), Enqueue(15), Enqueue(30), Dequeue(), Dequeue(15), Enqueue(52), Enqueue(52), Dequeue(), Dequeue(), Dequeue(), Dequeue.

| Operation   | Queue State.       |
|-------------|--------------------|
| Enqueue(2)  | 2                  |
| Enqueue(67) | 2, 67              |
| Enqueue(89) | 2, 67, 89          |
| Dequeue()   | 67, 89             |
| Enqueue(15) | 67, 89, 15         |
| Enqueue(30) | 67, 89, 15, 30     |
| Enqueue(52) | 67, 89, 15, 30, 52 |
| Dequeue()   | 89, 15, 30, 52     |
| Dequeue()   | 15, 30, 52         |
| Dequeue()   | 30, 52             |
| Dequeue()   |                    |

b) 62 + 63 \* -

|   |        |          |
|---|--------|----------|
| 6 | Push   | 6        |
| 2 | Push   | 6, 2     |
| + | +      | 8        |
| 6 | Push   | 8, 3     |
| 3 | * Push | 8, 6, 3. |
| * | *      | 8, 18    |
| . | -      | -10.     |
| - |        |          |